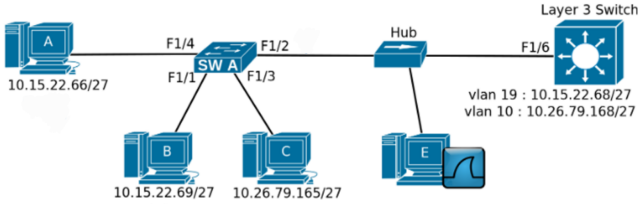
Tempo restante 0:46:57



Considering the above network where all PCs have the indicated IPv4 addresses and the correct gateway configured. Layer 3 switch has the port connected to the hub configured as trunk/inter-switch. The layer 3 switch has two virtual interfaces associated to VLAN19 and VLAN10. Switch A has the port connected to the hub configured as trunk/inter-switch. There are connectivity between all terminals. PC E is capturing packets on the link between the switches. Answer True or False to the following sentences:

After performing PINGs from PC C to 10.26.79.168, PC E will capture ICMP packets with a 802.1Q VLAN tag equal to 10.

T PC C → Vlan 10

PC A and PC C are connected to Switch A using access ports associated with the same VLAN.

F : Vlan 19 ← A, B 10.15.22.68 /26 → Endereço da rede é o 10.15.22.64 /26

64 + 32 = 96 − 2 = 94 (Endereço rede + Broadcast)

32 → 2⁵ = 32 → 26 / 05

Hosts disponíveis (10.15.22.65 → 10.15.22.94)

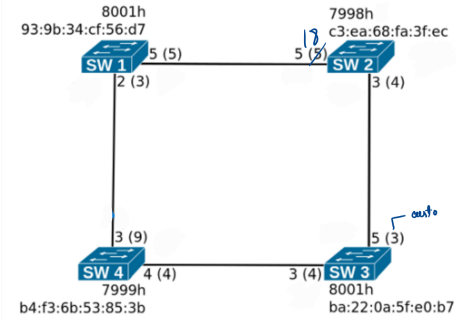
After performing PINGs from PC A to PC C, PC E will capture ICMP packets only with a 802.1Q VLAN tag equal to 19.

F : Vai capturar também com a tag 10 (PC A)

After sending one ICMP Echo Request packet from PC C to 10.15.22.69, PC E will capture only one ICMP Echo Request packet.

T PC C → PC B

??



RB — *Root Bridge*: Não é uma porta, mas sim o switch que serve de referência para todos a topologia STP. Todas as outras portas/switches calculam o caminho mais curto até o *Root Bridge*.

DP — *Root Port*: É a porta que tem o melhor caminho para encaminhar tráfego para fora de um segmento de rede. Em cada segmento, apenas uma porta é eleita como *Designated Port* para evitar *loops*.

DP — *Designated Port*: É a porta que tem o melhor caminho para encaminhar tráfego para fora de um segmento de rede. Em cada segmento, apenas uma porta é eleita como *Designated Port* para evitar *loops*.

Considering the above network where all switches have Spanning-Tree active. The respective switch MAC addresses and BPDU priorities (only the default value) and the STP port costs are between parenthesis. Answer True or False to the following sentences:

Changing the STP cost of port 5 of Switch 2 to 18, the Spanning-Tree will remain the same.

F

Changing the STP cost of port 5 of Switch 3 to 10, port 3 of Switch 4 will remain in the same state.

Changing the STP priority of Switch 2 to 7999h will produce no change on the state of all switches' ports.

Changing the STP cost of port 5 of Switch 1 to 12, port 3 of Switch 4 will remain in the same state.

Consider a network where all routers have OSPF properly configured and all interfaces have OSPF cost equal to 1. One of the routers has the following routing table:

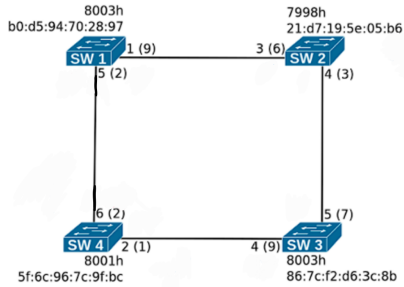
```
   10.0.0.0/8 is variably subnetted, 7 subnets, 1 mask
C       10.8.81.0/24 is directly connected, FastEthernet1/0
C       10.15.63.0/24 is directly connected, FastEthernet1/1
O E2    10.22.76.0/24 [110/4] via 10.8.81.10, 00:00:29, FastEthernet1/0
O E2    10.23.82.0/24 [110/4] via 10.8.81.6, 00:00:59, FastEthernet1/0
O E2    10.25.86.0/24 [110/4] via 10.15.63.10, 00:00:56, FastEthernet1/1
C       10.27.67.0/24 is directly connected, FastEthernet2/0
O E1    10.37.72.0/24 [110/2] via 10.15.63.3, 00:00:04, FastEthernet1/1
```

The OSPF domain has a single area and no neighbor router is announcing a default route. F

The OSPF domain has a single area and a neighbor router is announcing a default route with metric 0. F

The OSPF domain has more than one area and no router is announcing a default route. V

No routes to external networks are beeing distributed into the OSPF process and no router is announcing a default route. V



Considering the above network where all switches have Spanning-Tree active. The respective switches MAC addresses and SPT priorities are close to each switch, and the SPT port costs are between parenthesis. Answer True or False to the following sentences:

Changing the STP priority of Switch 2 to 7999h will produce no change on the state of all switches' ports.

Changing the STP cost of port 1 of Switch 1 to 11, port 5 of Switch 1 will change state.

Changing the STP cost of port 3 of Switch 2 to 26, the Spanning-Tree will change.

Changing the STP cost of port 5 of Switch 3 to 8, port 5 of Switch 1 will change state.